

# Julie Ha

## Contact Information

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## Education

**Boston University Computing and Data Sciences** Boston, MA

*Ph.D. Candidate*

Advisor: Mayank Varia

Research Interests: applied cryptography, computer science and law, usable security and privacy

**Boston University** Boston, MA

*B.A. in Computer Science with Honors*

Recipient of Computer Science Award for Academic Excellence

Senior Thesis: Oblivious Searchable Encryption for Irises

September 2018 - May 2022

## Publications

[1] R. Canetti, J. Ha, G. Kaptchuk

Analyzing Cryptography in Context: A Cryptography-Native Approach to Threat Modeling  
USENIX Security 2026

- Contribute a framework for analyzing sociotechnical goals of cryptographic deployments

[2] J. Ha, C. Cachet, L. Demarest, S. Ahmad, B. Fuller

Private Eyes: Zero-Leakage Iris Searchable Encryption.

ACM Conference on Data and Application Security and Privacy (CODASPY)

<https://eprint.iacr.org/2023/736>

- Contribute the first *zero-leakage* biometric database using locality sensitive hashes (LSHs) and oblivious maps
- Implemented Private Eyes in Python and tested implementation on real, synthetic, and random irises

[3] Y. Gvili, J. Ha, S. Scheffler, M. Varia, Z. Yang, and X. Zhang

TurboIKOS: Improved Non-interactive Zero Knowledge and Post-Quantum Signatures.

International Conference on Applied Cryptography and Network Security (ACNS) 2021

<https://eprint.iacr.org/2021/478>

- Contribute a new zero-knowledge argument proof in the MPC-in-the-head paradigm (MPCitH) that retains all benefits of MPCitH while providing concretely smaller proof sizes
- Implemented TurboIKOS in Python

## Works in Progress

[1] Measuring Content Transformation for Fair Use

- Contribute a formal technical definition of a fair-use property using elements of Kolmogorov complexity

[2] Digital Security and Privacy in Immigration Rights Organization

- Interview personnel at immigration-rights organizations to identify digital security and privacy concerns related to their use of existing digital tools

[3] Private Borrowing and Calendar Appointment System

- Building a cryptographically secure system for borrowing systems, such as libraries and bike shares, and calendar appointment systems, that prioritizes user privacy

## Awards

CDS Teaching Excellence Award (May 2026)

Hariri Institute Community Recognition Awards - Graduate Student Excellence Award (June 2023)

## Teaching Experience

Fall 2025: Graduate TA for DS653: Crypto for DS

Spring 2025: Graduate TA for DS593: Privacy in Practice

Spring 2023: Graduate TA for CS558: Network Security

Spring 2022: Grader for CS538: Fundamentals of Cryptography, Grader for CS558: Network Security

Fall 2021: Undergraduate Teaching Assistant for CS357: Introduction to Information Security

## Talks

**Building a Guidebook for Collaborations between Cryptographers and Social Good Organizations**

*Annual Boston Security Usability Research Day (ABSURD) (February 2026)*

**Private Eyes: Zero-Leakage Iris Searchable Encryption**

*ACM Conference on Data and Application Security and Privacy (June 2025)*

*University of California, Berkeley Security Seminar (May 2025)*

**TurboIKOS: Improved Non-interactive Zero Knowledge and Post-Quantum Signatures**

*University of Connecticut Security Seminar (July 2021)*

*International Conference for Applied Cryptography and Network Security (June 2021)*

## Posters

**Cryptography in Context: The Case Study of Apple's CSAM Scanning Proposal**

*New England Security Day 2024*

*Annual Boston Security Usability Research Day (ABSURD) 2024*

## External Reviewer

Privacy Enhancing Technologies Symposium (PETS) 2026

USENIX Security 2025

CCS 2024

USENIX Security 2023

CCS 2023

USENIX Security 2022

Crypto 2022

## Outreach

**CodeBreakers** Boston, MA (Remote)

Coordinator

- Instructed cryptography portion of a summer program for high school sophomores and juniors to help create relationships in STEM and introduce topics in computer science
- Created activities related to computer science to facilitate engagement remotely

June 2020 - August 2020

## References

Available upon request